

GTU Fundamental of AI - Question Bank (Based on Atul Prakashan Book)

Unit 1: Introduction to Artificial Intelligence

Weightage: ~30%

- Short/Long Questions:

1. Define Artificial Intelligence. Explain with suitable examples.
2. What are the three types of AI: ANI, AGI, ASI?
3. Explain the concept of intelligent agents.
4. Describe rationality and nature of environments.
5. How do intelligent agents work in different environments?
6. Explain characteristics of intelligent agents.
7. Differentiate between deterministic and stochastic environments.
8. Differentiate between static and dynamic environments.

Unit 2: Knowledge Representation

Weightage: ~20%

- Short/Long Questions:

9. Define knowledge representation. Why is it important in AI?
10. Explain different approaches to knowledge representation.
11. Describe Natural Language Processing (NLP) with examples.
12. What is pattern recognition? Explain its types.
13. Define expert system and mention its key components.

Unit 3: Basics of Machine Learning

Weightage: ~20%

- Short/Long Questions:

14. What is machine learning? How is it different from traditional programming?
15. Explain types of learning: supervised, unsupervised, reinforcement.
16. Describe Linear Regression, Logistic Regression, and SVM models.
17. What are the main goals of reinforcement learning?

18. What is the concept of “learning from examples”?

Unit 4: Deep Learning

Weightage: ~15%

- Short/Long Questions:

19. What is deep learning? How is it different from machine learning?

20. Describe the architecture of a typical neural network.

21. Explain CNN and RNN with differences.

22. What is backpropagation and its role in training?

23. Applications of deep learning in healthcare, finance, etc.

Unit 5: Modern AI

Weightage: ~15%

- Short/Long Questions:

24. What are Large Language Models (LLMs)? Give examples.

25. Case study: Applications of ChatGPT, Gemini, Bhashini, Krutrim.

26. What are current issues in deploying AI in real-world?

27. Future challenges in AI development.

28. What is explainability in AI?